AKSHAY KANGUDE

Robotics Engineer | ROS2 | Quadruped Locomotion | SLAM & Perception | Motion Control

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PROFILE SUMMARY

Robotics & Mechatronics Engineer with 3.5+ years of industrial experience across robotics, motion control, and IIoT automation. Specialises in quadruped locomotion (MIT-mode impedance control, FK/IK), ROS2-based autonomous systems, SLAM-based navigation, and OpenCV perception pipelines. Proven track record of deploying production-grade robotic systems end-to-end - from algorithm design and embedded control through field validation and real-time debugging. Equally at home with legged robot gait development, AMR navigation, multi-axis servo systems, and PLC/IIoT integration. First-principles problem solver with strong ownership across perception, planning, and actuation stacks.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Robotics & Autonomy: ROS2, Nav2, SLAM (LiDAR/Visual), Motion Planning, FK/IK, Trajectory Generation, URDF, Gazebo, RViz, Ignition

Legged Robotics: Quadruped kinematics (FK/IK), MIT Cheetah-style impedance control, stiffness and damping tuning (Kp/Kd), trot/walk/run gait generation, stance & swing phase control, whole-body dynamics, joint-space & task-space control

Perception & CV: OpenCV, LiDAR SLAM, RGB-D Camera, ToF & Ultrasonic sensing, visual servoing, human detection & tracking, depth estimation, multi-modal sensor fusion (IMU/LiDAR/RGB-D/ToF/Ultrasonic/Proximity), point cloud processing

Systems & Software: Real-time systems, multithreading, IIoT, TCP/IP, distributed systems, debugging & performance optimisation

Embedded & Control: Closed-loop control, MIT impedance controller, multi-axis servo systems, motion profiles, drive tuning, pneumatic valve control, DC actuator drive, DC servo motor control, limit switch integration, embedded C/C++

Microcontrollers & Electronics: ESP32, Arduino ATmega (Industrial), Arduino Uno R4, Raspberry Pi 4, relay driver circuits, optocoupler isolation stages, motor driver interfaces, I/O signal conditioning, high-current actuator interfacing

Protocols: CAN, EtherCAT, TCP/IP, RS485, I2C, Modbus, Ethernet

Tools & Platforms: VS Code, Git, GitHub, TIA Portal, FT Optix, Arduino IDE, SolidWorks, Inventor, AutoCAD

WORK EXPERIENCE

Motion Control Engineer – Assistant Manager - Bharat Forge – KSMS *July 2025 – Present*

Quadruped Robotics - MIT-Mode Locomotion & Aiming System

- FK/IK - 12-DOF Quadruped Kinematics
  - Sagittal (XZ) and lateral (Y) plane leg positioning; stable stance geometry via analytical IK solver
- MIT Impedance Control - Stiffness and Damping Tuning
  - Kp/Kd tuned per-joint across all 12 joints; compliant, terrain-adaptive contact forces
- Gait Library - Walk, Trot, Run
  - Stance/swing scheduling, foot-trajectory generation, diagonal pair sync, dynamic bounding with flight phase
- Whole-Body Execution Pipeline
  - Joint-space torque commands with IMU and encoder feedback; safety limits for high-speed locomotion
- Joint Torque Balancing - Standing and Moving
  - Even GRF distribution for static equilibrium; dynamic torque redistribution to prevent tip-over and joint overload
- Body Orientation Control - IMU-Based
  - Real-time pitch, roll and yaw compensation; torso levelling on uneven terrain in standing and moving modes
- LiDAR-SLAM and Navigation - ROS2/Nav2
  - Autonomous map building, localisation and obstacle-aware path planning in unstructured environments
- Multi-Modal Sensor Fusion
  - ToF, Ultrasonic, Proximity, Limit Switches - close-range detection and joint end-stop protection

    - RGB-D Camera - depth-augmented perception; LiDAR - global mapping; unified environmental awareness layer
- Actuation Stack - Pneumatic Valves, DC Motors, DC Servos
    - Valve timing, pressure sequencing, DC motor drive and closed-loop servo position/velocity control
- Aiming and Targeting System - OpenCV
    - Human detection, object tracking, depth estimation, closed-loop visual servoing for body/sensor-head orientation
- Unified Perception and Locomotion Loop
    - Camera, LiDAR and state estimation fused in real time; latency, noise and sync optimised for field deployment
    - Tools: ROS2, MIT Controller, OpenCV, LiDAR SLAM, Python, C++, IMU, CAN bus | Sensors: ToF, Ultrasonic, Proximity, Limit Switches, RGB-D, LiDAR | Actuators: Pneumatic Valves, DC Motors, DC Servos | MCUs: ESP32, ATmega, Uno R4, RPi 4 | Electronics: Relay Drivers, Optocouplers, Motor Drivers

AMR & Industrial Robotics

- ROS2 AMR and Forklift Deployment
    - Motion control and navigation systems deployed in live industrial environments
- Closed-Loop Multi-Axis Control
    - Encoder integration, drive tuning and real-time feedback optimisation across multi-axis platforms
- Synchronised Multi-Drive Motion Profiles
    - Coordinated drive profiles under hard real-time constraints
- Sensor and Protocol Integration
    - CAN, EtherCAT, TCP/IP - sensors and actuators into scalable production-grade robotic architectures
- System-Level Debugging - Live Deployments
    - Resolved issues across control, communication and hardware layers during active industrial operation
- Full-Stack Contribution
    - Embedded control, mechanical integration, system validation - prototype through production
- Embedded Electronics Subsystems
    - ESP32, Arduino ATmega Industrial, Uno R4, RPi 4 - relay drivers, optocouplers, motor driver interfaces for noise-immune high-current I/O control

IIoT & Automation Engineer - ENLIVE AUTOMATION S.R.L – Italy *November 2024 – June 2025*

- Scalable IIoT Systems
    - Real-time monitoring & control of multiple industrial controllers across distributed plants
- Distributed Communication Architecture
    - Python-based multi-controller comms with centralised servers; web HMI via FT Optix
- PLC / SCADA / Robotics Integration
    - End-to-end automation solutions from design through production deployment
- Distributed Control Optimisation
    - Managed real-time data flow, communication latency & multi-device synchronisation

Application Engineer - Trio Motion Technology India PVT LTD *July 2020 – February 2022*

- Multi-Axis Motion Control Applications
    - Real-time control for conveyors, packaging and material handling systems
- VFFS and HFFS Machine Programming and HMI Development
    - Programmed Vertical Form Fill Seal (VFFS) and Horizontal Form Fill Seal (HFFS) packaging machines using Trio motion controllers
    - Developed full HMI interfaces for machine operation, parameter setting, fault display and production monitoring
    - Implemented real-time production data fetching and live display on web portals for remote monitoring and reporting
- Commissioning & On-Site Debugging
    - Motion controllers, servo drives & robotic systems tested & deployed at client sites
- Diagnostics Tooling
    - Built testing tools for motion controllers & servo drives; improved validation speed
- Hardware-Software Integration
    - Resolved issues across sensors, motors and communication systems
- Electronics Testing Kit - Full Ownership, Sole Engineer

  - End-to-end ownership: component selection, mechanical CAD, manufacturing, electronics wiring and system integration - entirely independently
  - Designed enclosure and panel layout in CAD; managed full manufacturing process including machining, assembly and mechanical fit-out
  - Selected all electronic components; designed and wired complete circuit including power, signal conditioning and interface boards
  - Covered all communication protocols: CAN, RS-485, RS-232, Digital I/O, Analog I/O, HMI CAN I/O cards
  - Enabled diagnosis of faulty controllers, servo drives, motors, encoders and I/O ports from a single test rig
  - Reduced fault diagnosis and testing time by 75% - adopted across factory acceptance testing and field commissioning

ACADEMIC & PERSONAL PROJECTS

PAL TIAGo Robot – Patient Analysis (Master Thesis) – *ROS2 | Nav2 | Gazebo | OpenCV | Ubuntu*

- Autonomous patient analysis robot - TIAGo + ROS2
  - Nav2 navigation, CV-based perception & behaviour logic; validated in Gazebo with real-world deployment design

LiDAR SLAM Simulator – *Python | pygame | NumPy | Bug2 Navigation*

- Full-featured LiDAR SLAM simulator in Python/pygame
  - Configurable FOV, Bug2 avoidance, real-time map building, collision detection, modular toolbar UI

Quadruped 3D Kinematics Simulation – *Python | matplotlib | NumPy*

- 3D animated 5-bar linkage leg - matplotlib
  - XZ sagittal kinematics, Z-axis K/H pivot, Y-axis C-carrier offset; FK/IK validated against analytical solution

Siemens S7-1200 PLC Dashboard – *Python | Flask | snap7 | WebSocket*

- Real-time PLC web dashboard - Python/Flask/snap7
  - 4-mode machine state, bypass/force panel, axis jog, sim/interlock architecture; dark-theme UI

Cutting & Sealing Machine – Motion Profile Design – *MATLAB | Inventor*

- Mechanical design + motion profile simulation
  - Torque/power calculations, trapezoidal motion curves in MATLAB/Inventor

Shoe Sanitation Machine (Final Year) – *Embedded C | SolidWorks | Arduino | Mobile App*

- End-to-end electromechanical system
  - Component selection, SolidWorks CAD, firmware, wiring & mobile app for device control

EDUCATION

M.Tech – Industrial Automation, Robotics & Mechatronics Engineering - University of Pavia, Italy*2022 – 2025*
Score: 96/110

B.Tech – Mechatronics Engineering - Symbiosis Skills & Professional University, Pune *2017 – 2021*
CGPA: 8.5/10

INTERNSHIPS

E-Internship - Sai Samarth Engineering, Goa

- Gantry & articulated robot design - SolidWorks / AutoCAD
  - Chocolate packaging automation & pick-and-place applications

Trainee Intern - Success Industries, Pune

- Mercedes-Benz oil filter assembly line automation
  - Mechanical design & production support across design, electrical & assembly teams

Open to Robotics, Mechatronics, Motion Control, Legged Robotics and Autonomous Systems roles globally